

Master's Comprehensive Examination

Part I

Theory (90 minutes)

November 20, 2004

Instructions: Answer all four questions. Open book and open notes.

1. Find the approximate probability that among 10,000 random digits the digit 7 appears not more than 968 times. Answer must be a number.

2. Let X be distributed according to a binomial distribution with parameters $n = 2$, $p = 1/2$. Let Y be binomial with parameters $n = 3$, $p = 1/3$. Assume X and Y are independent.

(a) Find $P\{X \geq 1\}$.

(b) Find $P\{X \geq 2|X \geq 1\}$.

(c) Find $P\{X = 2, Y = 3\}$.

(d) Find $P\{X = 2|Y = 3\}$.

(e) Make a two-way table whose entries are the joint probability distribution of X, Y , i.e. $P\{X = x, Y = y\}$, $x = 0, 1, 2$; $y = 0, 1, 2, 3$.

(f) Find $P(X = 1|X + Y = 2)$.

3. Let X_1, X_2, X_3 be independent normal variables with mean μ and variance σ^2 . Both μ and σ^2 are unknown.

(a) Find an unbiased estimator of $\mu + 2$.

(b) Find an unbiased estimator of $2\mu + 3\sigma^2$.

4. Let X_1, X_2 be independent $N(\mu, 4)$. To test $H_0 : \mu = 0$ vs $H_1 : \mu > 0$ the UMP test of size $\alpha = .05$ is reject if $\bar{X} > C_\alpha$. Find C_α and find the power of the test when $\mu = 1$.